

Soc Internship day 22

1. Explain MITRE ATT&CK

MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) is a knowledge base and framework that documents real-world cyberattack behaviors. It helps security teams understand how attackers gain access, move through networks, steal data, and achieve their objectives.

It is widely used for:

- Threat hunting
- Security monitoring
- Incident response
- Red team exercises
- Detection engineering

2. Define Tactics and Techniques

Tactics

Tactics describe the attacker's goal during a stage of an attack.

Examples:

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery
- Lateral Movement
- Collection
- Exfiltration
- Impact

Techniques

Techniques describe how the attacker achieves a tactic.

Examples:

- Phishing (Initial Access)
- PowerShell (Execution)
- Credential Dumping (Credential Access)
- Remote Services (Lateral Movement)
- Data Encrypted for Impact (Ransomware)

3. Analyze a Sample Attack

Scenario: Ransomware Attack

Attack Stage	MITRE Tactic	Example Technique
Employee receives malicious email	Initial Access	Phishing
User opens attachment	Execution	User Execution
Malware installs itself	Persistence	Registry Run Keys
Malware disables antivirus	Defense Evasion	Impair Defenses
Passwords are stolen	Credential Access	Credential Dumping
Attacker scans network	Discovery	Network Service Scanning
Attacker spreads to servers	Lateral Movement	Remote Services
Files are collected	Collection	Data from Local System
Data is uploaded to attacker	Exfiltration	Exfiltration Over Web Service
Files are encrypted	Impact	Data Encrypted for Impact

Key Findings

- **Entry Point:** Phishing email.
- **Main Risk:** Credential theft and ransomware deployment.
- **Affected Assets:** User PCs, servers, and sensitive data.
- **Possible Indicators of Compromise (IoCs):**
 - Suspicious email attachments
 - Unusual PowerShell activity
 - Multiple failed logins
 - Large outbound data transfers
 - Sudden file encryption

Recommended Defenses

- Email filtering and anti-phishing controls
- Multi-Factor Authentication (MFA)
- Endpoint Detection and Response (EDR)
- Network segmentation
- Regular backups
- Security awareness training

MITRE ATT&CK helps defenders map each stage of this attack to known attacker behaviors, making detection and response more effective.